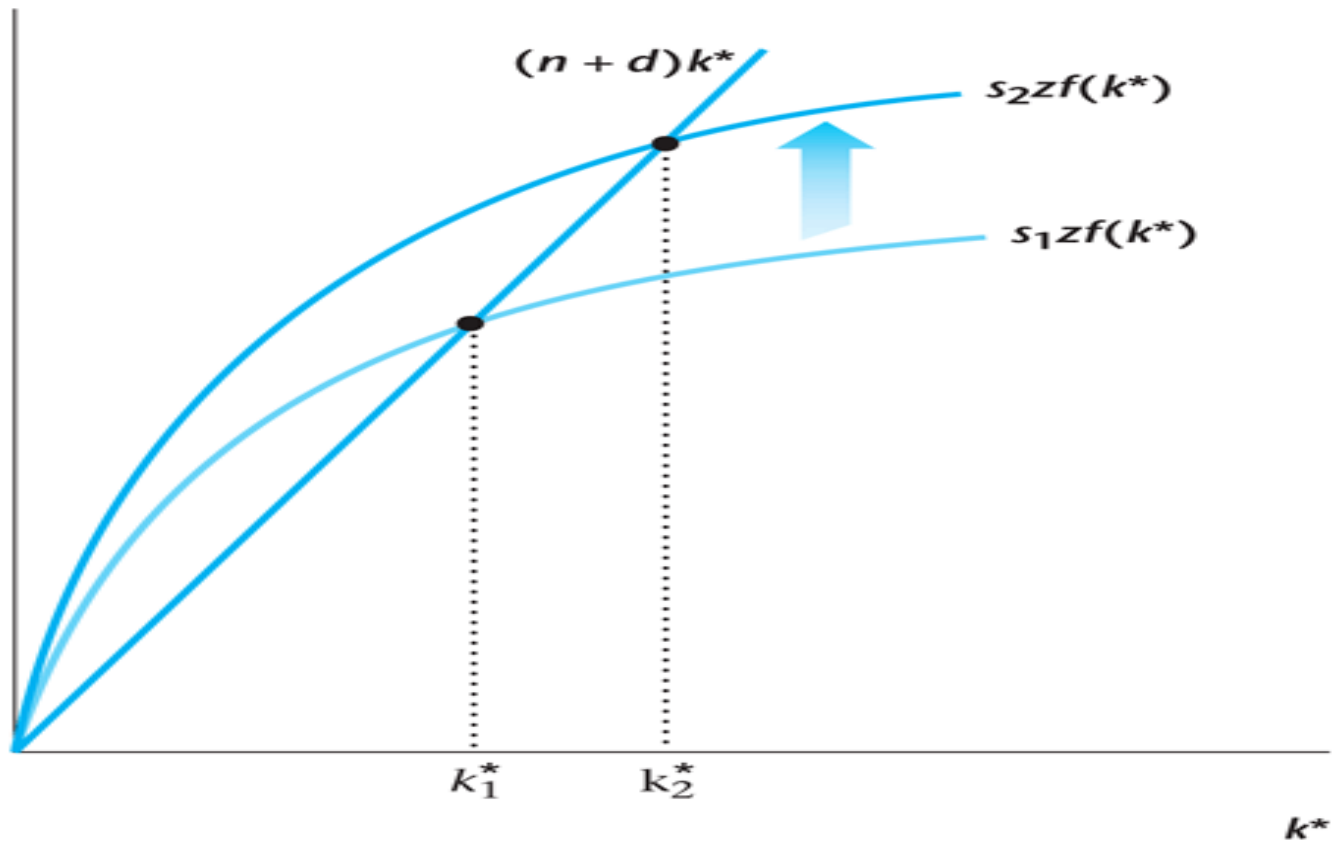


International Economic Development

ECON 4660/8666

Lecture 8 Capital and Saving

Saving and growth



Inter-temporal choices

The consumer's current-period budget constraint:

$$c + s = y - t$$

The consumer's future-period budget constraint:

$$c' = y' - t' + (1 + r)s$$

Consumer's Lifetime Budget Constraint

- Substitute in the current-period budget constraint obtaining lifetime budget constraint:

$$c + \frac{c'}{1+r} = y - t + \frac{y' - t'}{1+r}$$

- $\frac{1}{1+r}$: relative price of future consumption
 - If I give up 1 unit of consumption today, I can consume $1 + r$ unit of consumption tomorrow.
 - If I give up $\frac{1}{1+r}$ unit of consumption today, I can consume 1 unit of consumption tomorrow.

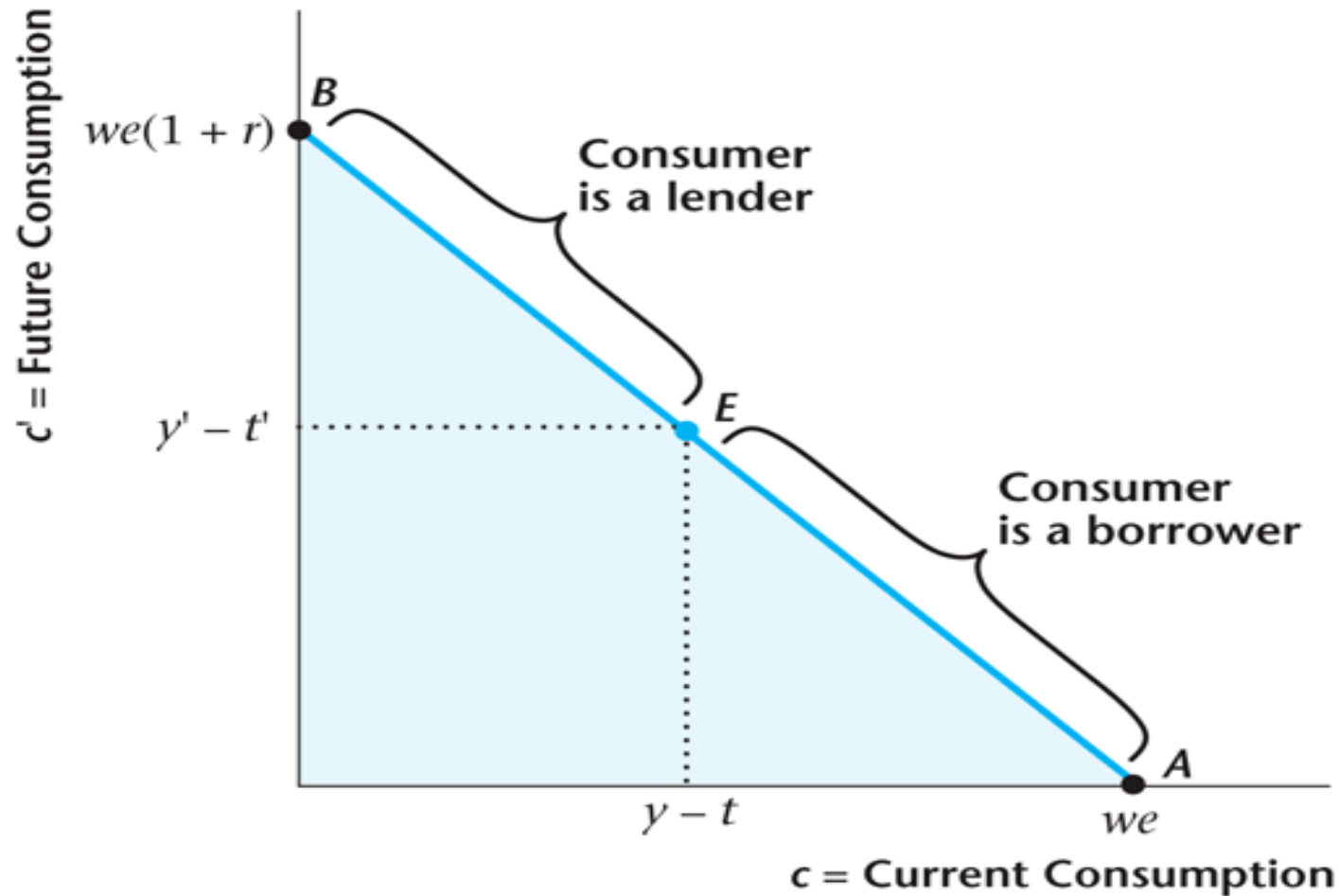
Simplified Lifetime Budget Constraint

$$we = y - t + \frac{y' - t'}{1 + r}$$

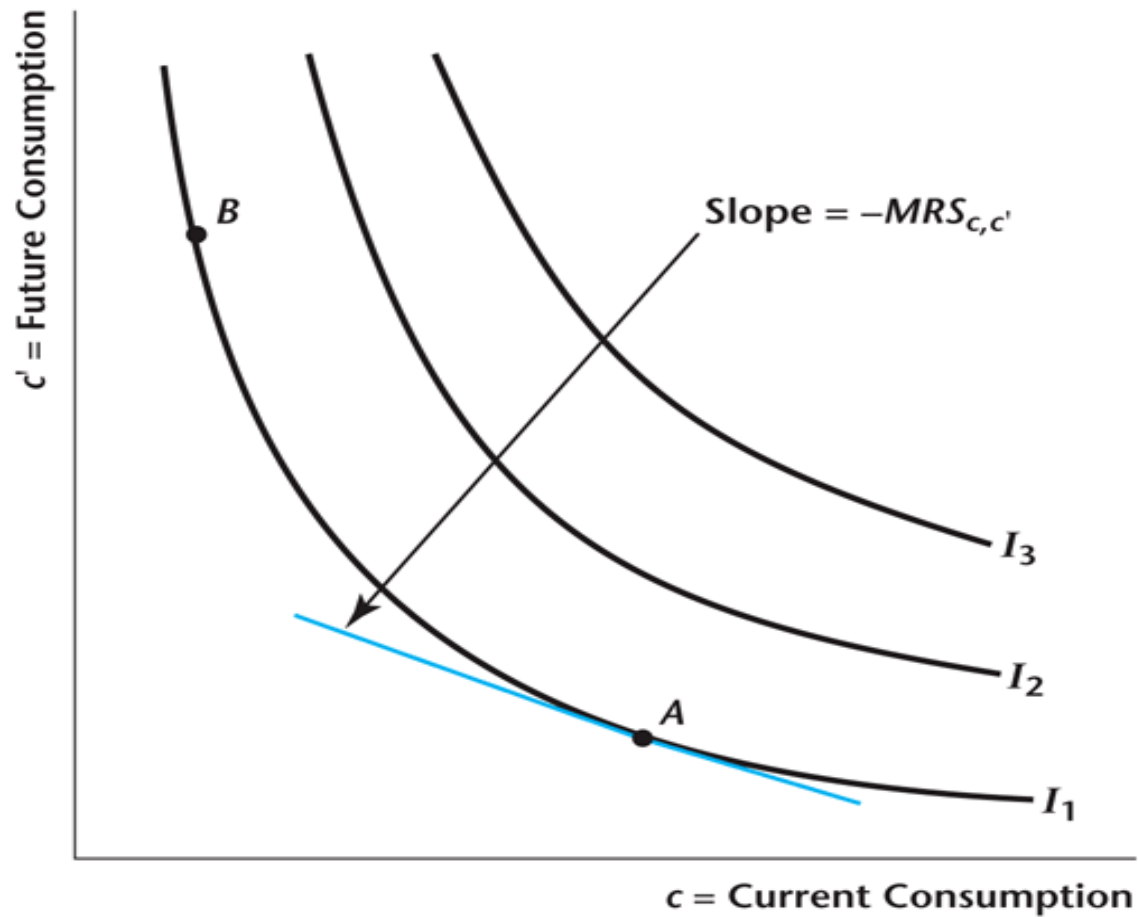
$$c + \frac{c'}{1 + r} = we$$

$$c' = -(1 + r)c + we(1 + r)$$

Consumer's Lifetime Budget Constraint



A Consumer's Indifference Curves



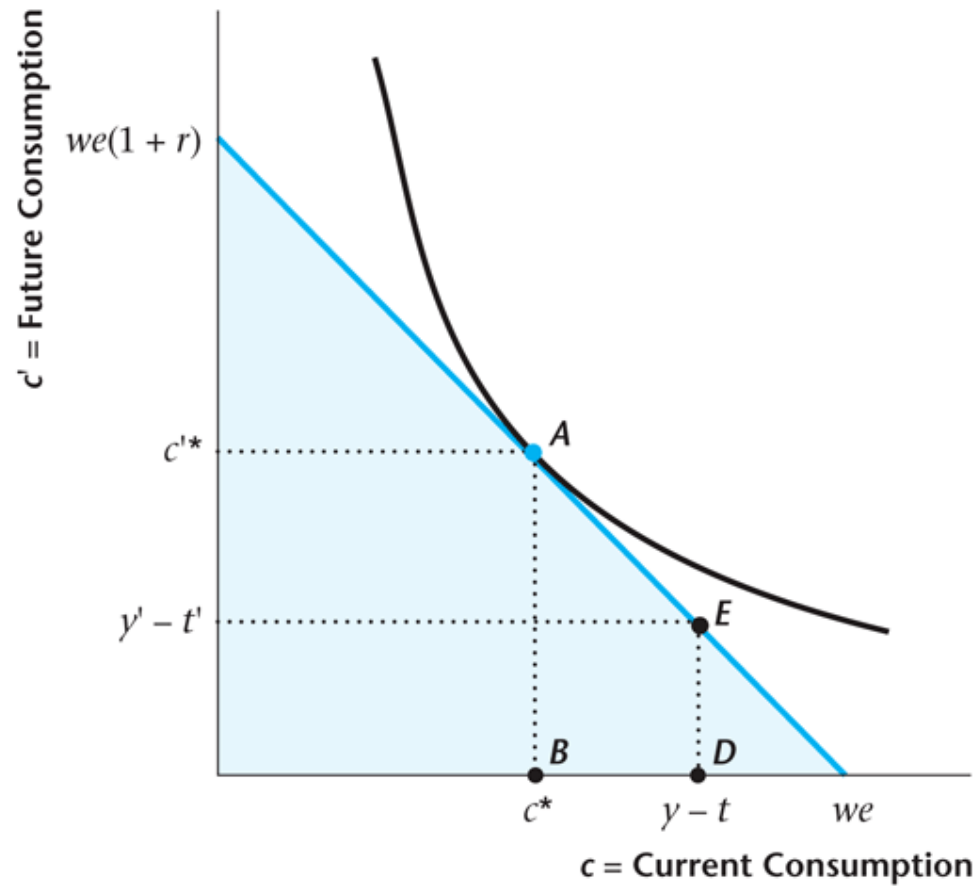
Optimization

- Marginal condition that holds when the consumer is optimizing:

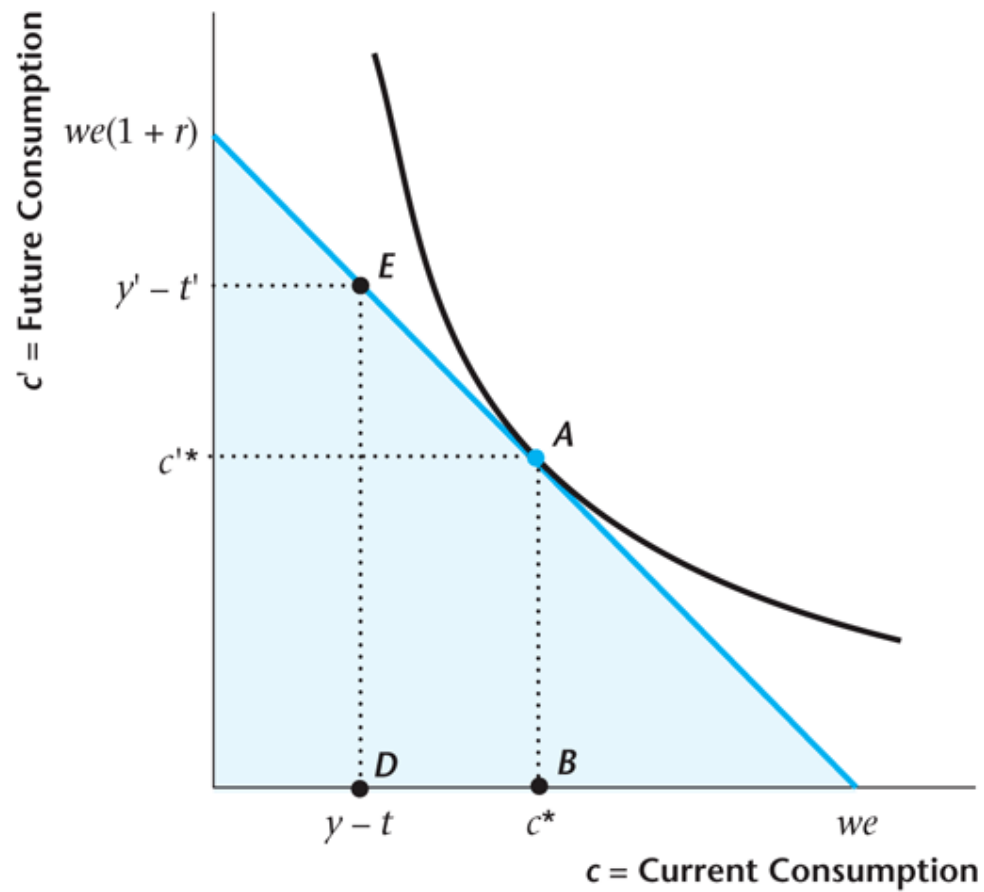
$$MRS_{c,c'} = 1 + r$$

- *Market*: 1 unit of c in exchange of $1 + r$ unit of c' .
- *Preferences*: 1 unit of c in exchange of $MRS_{c,c'}$ unit of c' .

A Consumer Who Is a Lender



A Consumer Who Is a Borrower



Inter-temporal choices

- Consumption Smoothing
 - The ability to borrow or lend creates a trade-off between current and future consumption.
 - c and c' depend only on lifetime wealth, w_e , not its composition.
- But saving does depend on composition of lifetime wealth:
 - Consumers with high first-period income are more likely to be savers (lenders).
 - Consumers with high second-period income are more likely to be borrowers, using future income to fund current consumption.
- Key assumption: Saving (borrowing) markets work perfectly.

Capital and Saving

Capital: machinery, equipment, tools,

Capital fundamentalism:

- Capital is the key to economic growth.

Capital and Saving

I. Investment and Growth

- Investment is positively related to the growth rate of per capita GDP.
- Ross Levine and David Renelt: American Economic Review, 1992. “A Sensitivity Analysis of Cross – Country Growth Regressions.”

- Harrod - Domar Model:

$$g = \frac{s}{ICOR}$$

g- growth rate,

s- saving rate,

ICOR- Incremental capital- output ratio.

The lower the ICOR is, the smaller the saving rate will be required for sustained growth.

Example:

Two countries, A and B, identical initially.

2000 Per capita income:\$400

Population: 2.5 million

Investment ratio: 15%, $\frac{I}{GDP}$

1990-2000 ICOR: 3.5

Annual growth rate: 4.3% ($= \frac{0.15}{3.5}$)

Policy changes

2000- A: capital-intensive investment ICOR: 4

Steel mills, oil refining

B: Labor-intensive investment ICOR: 3

Shoe manufacturing

- Annual growth rate of GDP:

$$g = \frac{S}{ICOR} = \frac{I}{ICOR} \quad \text{GDP} = 1000$$

A: ICOR = 4, assume that the growth rate of investment = 5%

$$2000: I = \frac{150}{1000} = 0.15, g = \frac{0.15}{4} = 3.75\%$$

$$2001: I = \frac{150 \times (1 + 5\%)}{1000 \times (1 + 3.75\%)} = \frac{157.5}{1037.5} = 0.1518, g = \frac{0.1518}{4} = 3.78\%$$

.....

$$2005: I = \frac{182.3}{1161.7} = 0.1569, g = 3.92\%$$

assume that the growth rate of investment = 10%

$$2000: I = \frac{150}{1000} = 0.15, g = \frac{0.15}{4} = 3.75\%$$

$$2001: I = \frac{150 \times (1 + 10\%)}{1037.5} = \frac{165}{1037.5} = 0.159, g = \frac{0.159}{4} = 3.98\%$$

.....

$$2005: I = \frac{219.6}{1174.1} = 0.187, g = \frac{0.187}{4} = 4.68\%$$

B: ICOR = 3, assume growth rate of investment = 5%

$$2000: I = \frac{150}{1000} = 0.15, g = \frac{0.15}{3} = 5\%$$

$$2001: I = \frac{157.5}{1050} = 0.15, g = \frac{0.15}{3} = 5\%$$

$$2005: I = \frac{182.3}{1215.5} = 0.15, g = 5\%$$

ICOR = 3, growth rate of investment = 10%

$$2000: I = \frac{150}{1000} = 0.15, g = \frac{0.15}{3} = 5\%$$

$$2001: I = \frac{165}{1050} = 0.157, g = \frac{0.157}{3} = 5.2\%$$

$$2005: I = \frac{219.6}{1232} = 0.178, g = \frac{0.178}{3} = 5.9\%$$

With lower ICOR, Country B grows faster even with lower growth rate of investment.

- Allocation of Investment Matters

Example 1:

Song, Storesletten and Zilibotti, AER 2011,

-- Reallocation of resource from low productivity State-owned firms to high productivity private firm explains the high output growth, sustained returns on capital

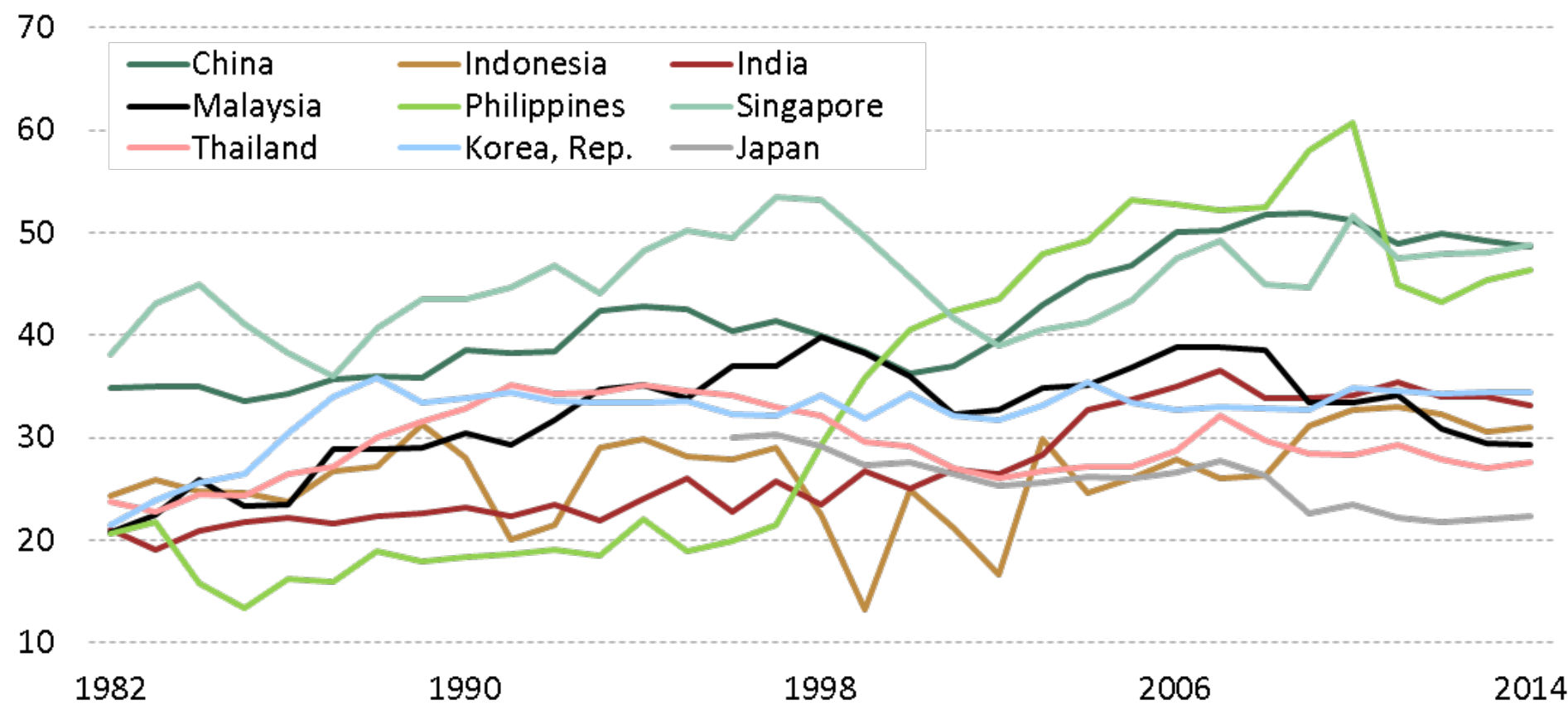
Example 2:

J. Bradford De Long and Lawrence Summers, QJE, 1991

-- Equipment investment is positively related to economic growth.

Chart 1: National Savings Rate Trend (Asian Countries)

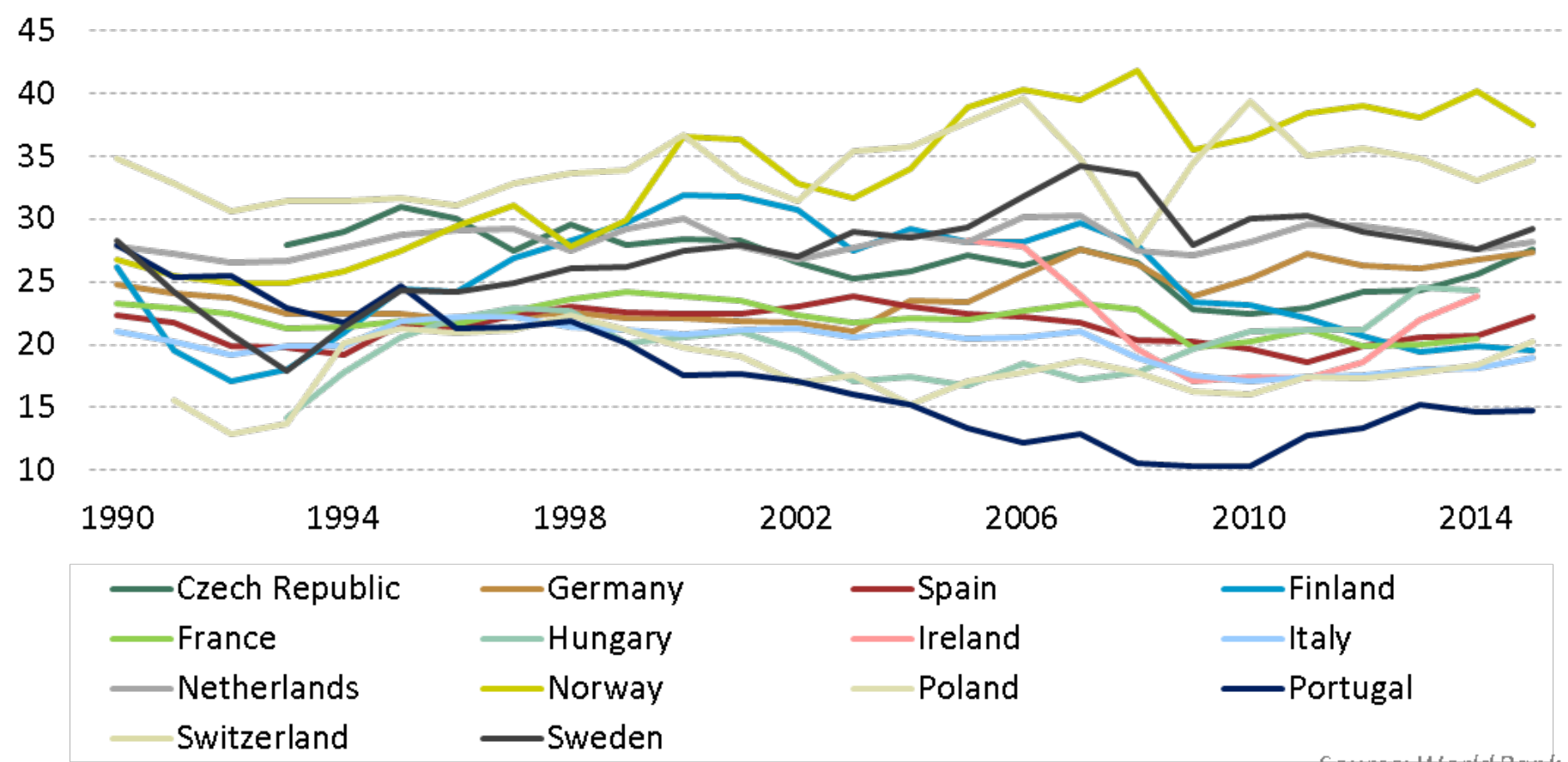
National Saving As % of GDP; As of 12/31/2014



Source: World Bank

Chart 2: National Savings Rate Trend (European Countries)

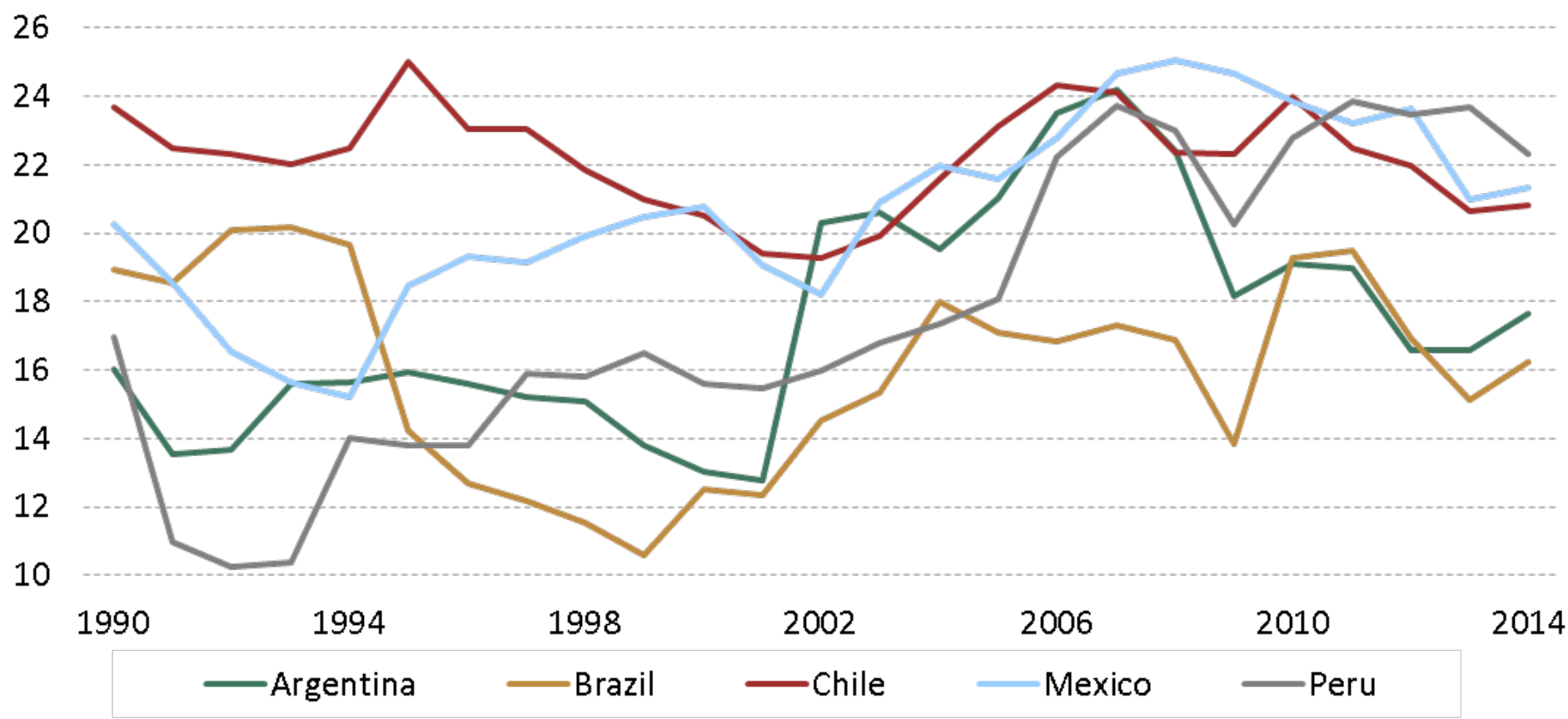
National Saving As % of GDP; As of 12/31/2014



Source: World Bank

Chart 3: National Savings Rate Trend (Latam Countries)

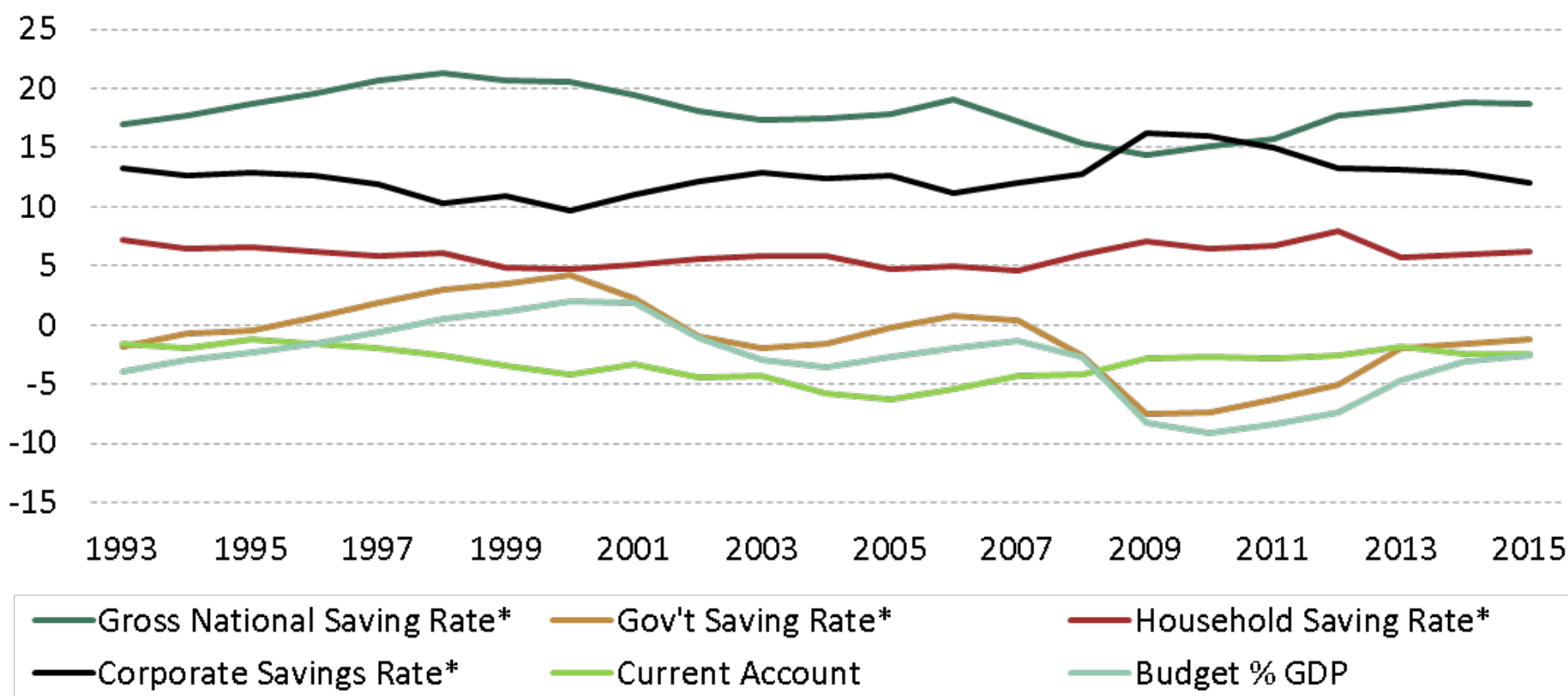
National Saving As % of GDP; As of 12/31/2014



Source: World Bank

Chart 9: United States Savings Rates

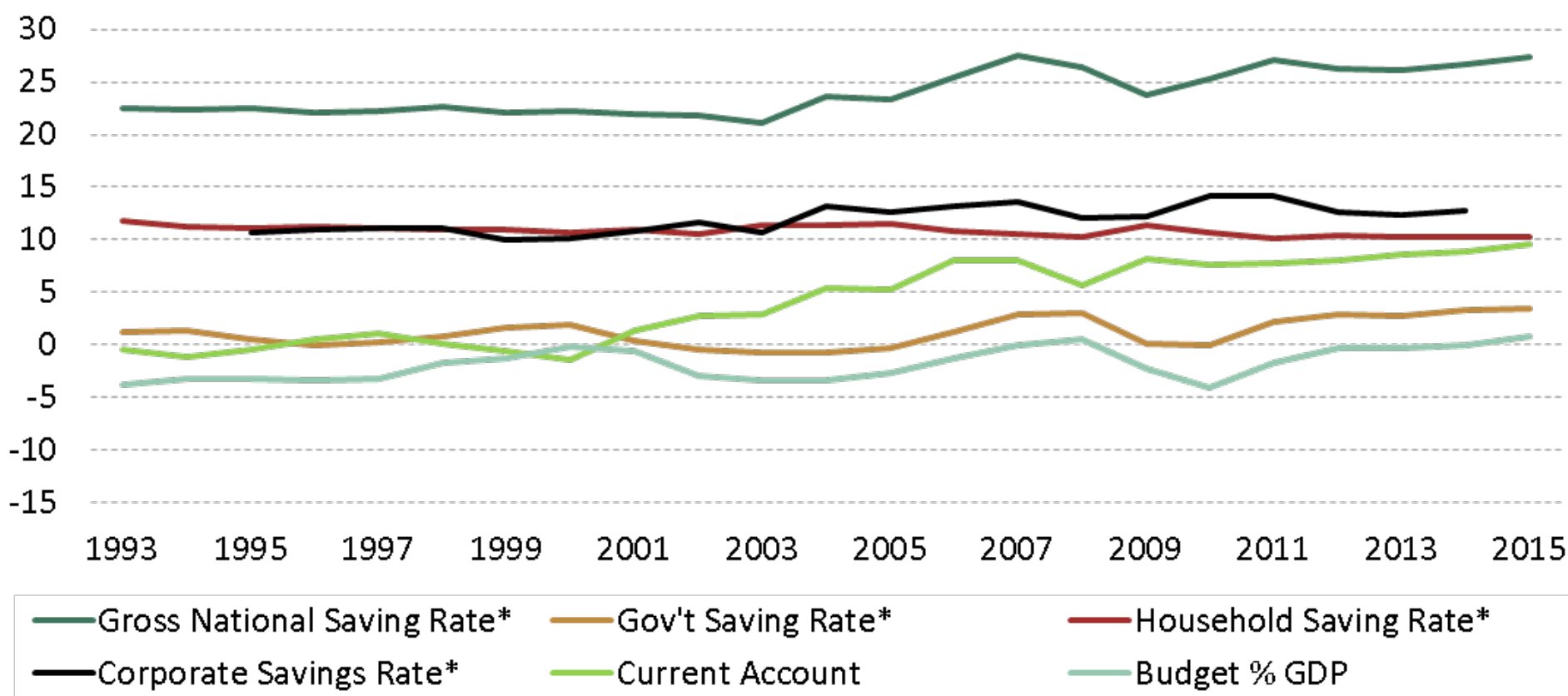
As of 12/31/2015



*As % of Nominal GDP
Source: Thomson Datastream

Chart 12: Germany Savings Rates

As of 12/31/2015



*As % of Nominal GDP
Source: Thomson Datastream

II. Sources of Savings

$$S = S_d + S_f$$

- S_d = domestic savings.
- S_f = foreign savings.

$$S_d = S_g + S_p$$

- S_g = government savings.
- S_p = private savings.

$$S_g = S_{gb} + S_{ge}$$

- S_{gb} = budgetary savings, which is equal to government revenues minus government expenditures (including military expenditures).
- S_{ge} = government enterprises' savings.

$$S_p = S_{pc} + S_{ph}$$

- S_{pc} = corporate savings, which is after-tax corporate income minus dividends paid to stockholders.
- S_{ph} = household savings, which is disposable income minus consumption.

$$S_f = S_{fo} + S_{fp}$$

- S_{fo} = foreign official savings, or foreign aid.
- S_{fp} = foreign private savings.

$$S_{fp} = S_{fpd} + S_{fpe}$$

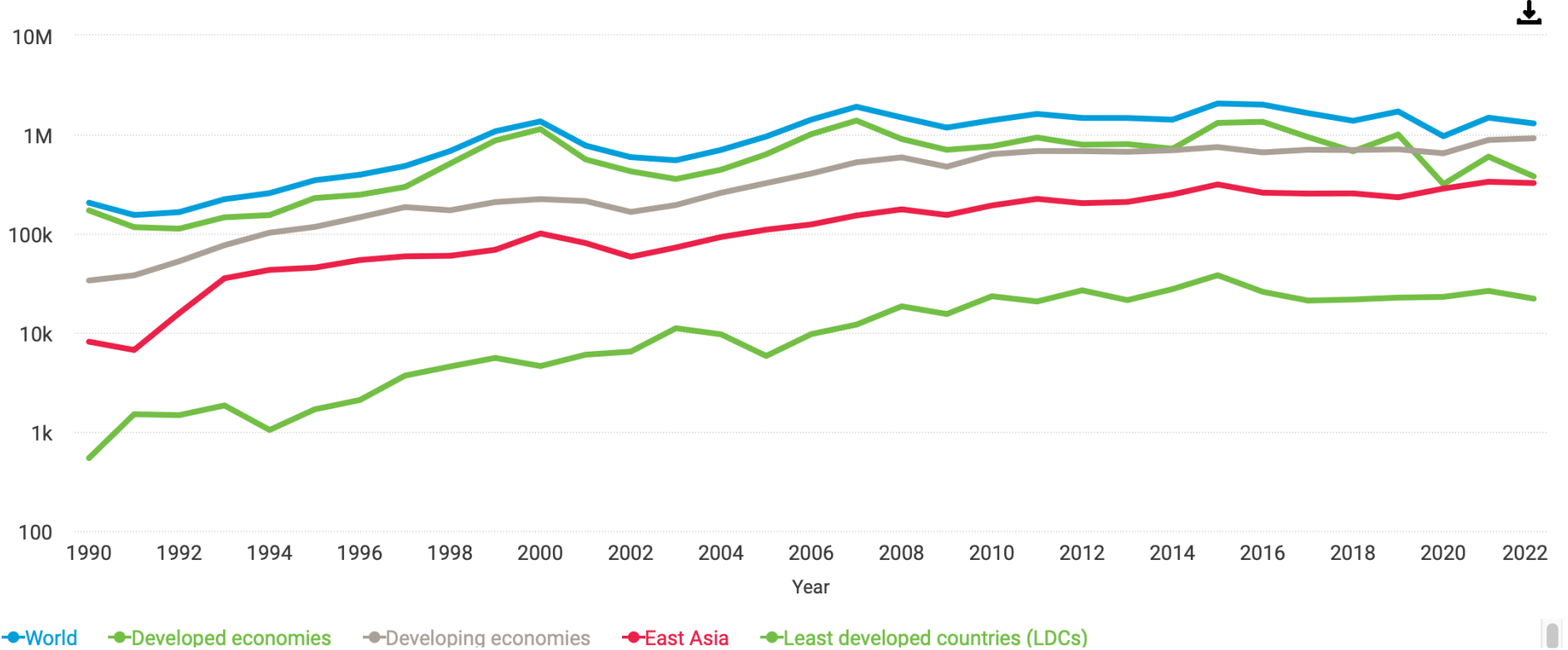
- S_{fpd} = external commercial borrowing, or debt finance.
- S_{fpe} = foreign direct investment, equity finance.

$$S = [(S_{gb} + S_{ge}) + (S_{pc} + S_{ph})] + (S_{fo} + S_{fpd} + S_{fpe})$$

By region and economy, 1990–2022

LINEAR LOG INFLOWS OUTFLOWS

Millions of dollars



By region and economy, 1990–2022

LINEAR LOG INFLOWS **OUTFLOWS**

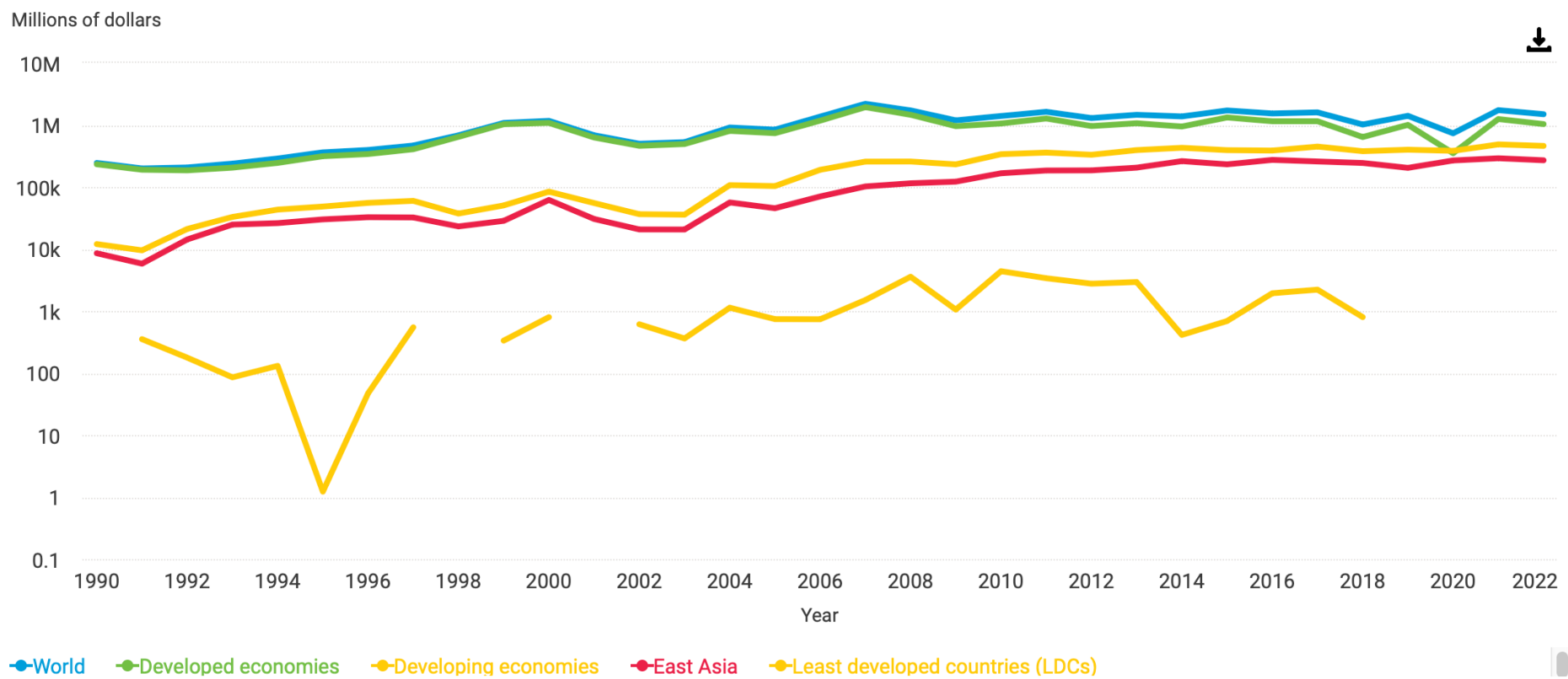
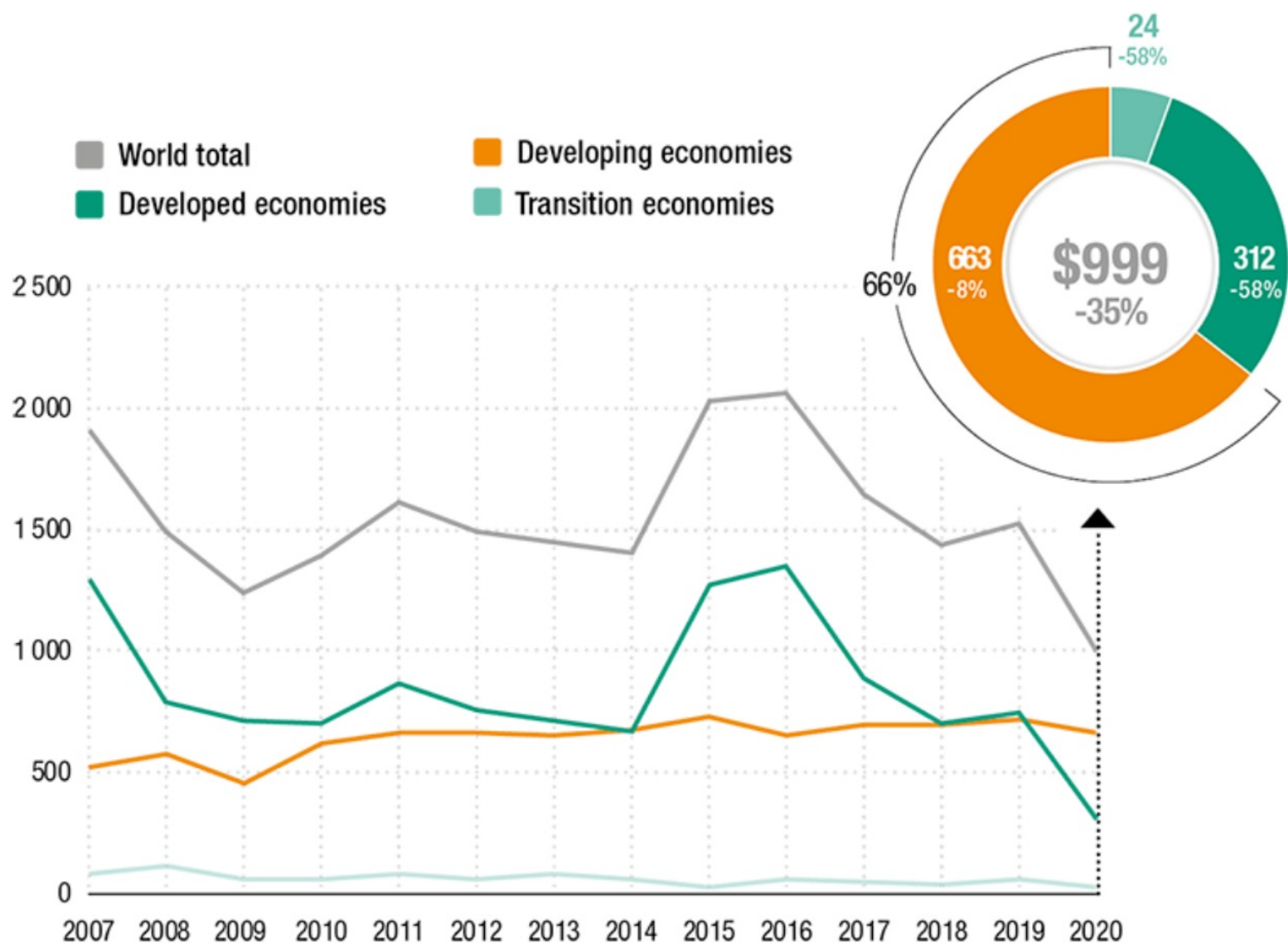


Figure 1 - Foreign direct investment inflows, global and by group of economies, 2007–2020
(Billions of dollars and per cent)



Source: UNCTAD, World Investment Report 2021.

III. Determinants of private savings.

- A. Household savings, some Stylized facts
 - High-income households tend to save larger fractions of their income in a given country.
 - Overtime, household savings ratios tend to be constant, particularly in industrial countries.
 - Across countries, savings ratios are not related to income.

Three models:

(1) Keynesian absolute – income hypothesis

$$S = a + sY^d$$

$$S = Y^d - C$$

S = savings,

a = constant, negative,

s = marginal propensity to save, $0 < s < 1$, $s = \frac{\Delta S}{\Delta Y^d}$

Y^d = current disposable income.

Example: $a = -100$,

$$s = 0.3,$$

If $Y^d = 1000$, then $S = -100 + 300 = 200$

$$APS = \frac{\text{Savings}}{\text{Income}} = \text{Saving ratio} = \frac{200}{1000} = 20\%$$

If $Y^d = 2000$, then $S = -100 + 600 = 500$

$$\text{Saving ratio} = \frac{500}{2000} = 25\%$$

- APS (or savings ratio) $\uparrow$ as income $\uparrow$.

This may be true in S.R. but not in L.R.

(2) The relative income hypothesis

James Duesenberry,

$$C_1 = a + (1-s) Y_1^d + bC_n$$

$$S_1 = Y_1^d - C_1$$

C_1 – consumption in period 1,

a – positive, constant,

$$MPC: c = 1 - s$$

Y_1^d – disposable income in period 1,

C_n – previous high level of consumption.

$$0 < s < 1, 0 < b < 1.$$

The SR consumption moves upward overtime,
“Ratchet effect”



(3) Permanent–Income Hypothesis

Milton Friedman, 1950s

$$S = a + b_1 Y_p + b_2 Y_t$$

S = savings, a = constant, Y_p = permanent income shock,
 Y_t = transitory income shock.

- Permanent income:
Yield from assets (physical and human capital),
stable over time
- Transitory income:
Unexpected income, changes in asset values,
lottery winnings, etc.
- Extreme version: $S = a + Y_t$
Modified version: $0 < b_1 < b_2 < 1$

(4) The Class Theory

Nicholas Kaldor

$$S = S_w L + S_c P$$

S_w – workers' saving propensity, L – labor income,

S_c – capitalists savings propensity,

P – property income (profits, interests, rents).

$$0 < S_w < S_c < 1$$

- This theory can explain fact (1) and fact (3) given that factor shares are different across countries.
- Other determinants of savings include: interest rate, age, structure of population, locations (rural save more than urban).

B. Corporate Savings

- Corporate savings as percentage of total income: 5% in industrial countries.
- Corporate Savings is relatively small in most developing countries since corporate sector is small.

IV. Determinants of foreign savings

A. Foreign Private Investment

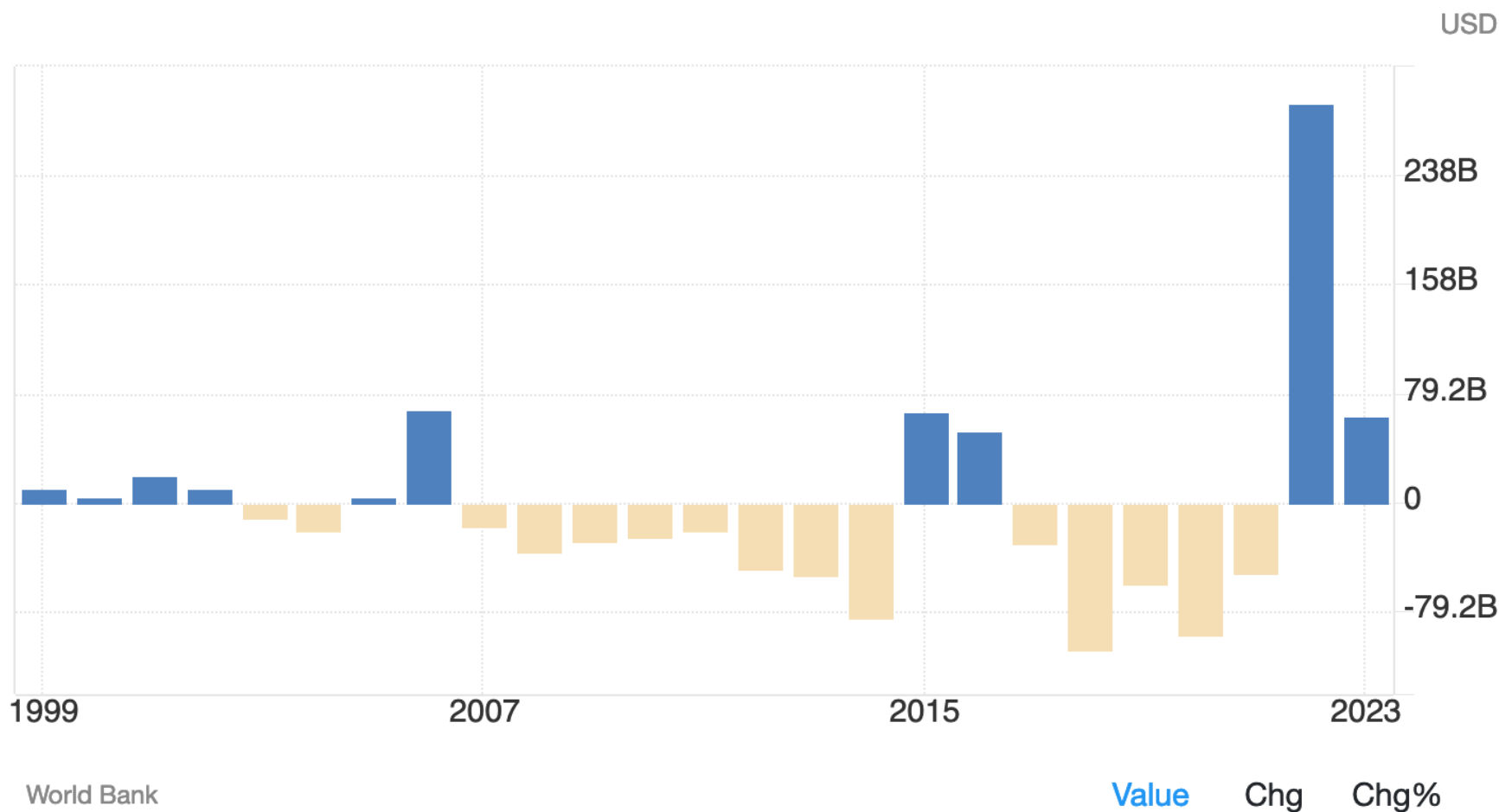
1. Portfolio investment : Purchases of stocks, bonds but not involving a controlling ownership (<10 %)

LDCs	\$4 billion	1990
------	-------------	------

	\$30 billion	1997
--	--------------	------

Less than 20% of FDI.

Portfolio Investment, net (BoP, current US\$) in China



2. Direct Investment (FDI):

- 1980s-1990s: This period saw a significant surge in FDI, especially to developing countries. The reasons included liberalization policies, the end of the Cold War, and the opening up of many large economies like China and India.
- 2000s: The early 2000s saw a continuation of the upward trend in FDI. However, the global financial crisis in 2007-2008 led to a sharp decline in FDI flows worldwide in 2008 and 2009.

2. Direct Investment (FDI):

- 2010s: Recovery and Surge (2010-2015): After the global financial crisis, there was a recovery in FDI, reaching new peaks around 2015.
- Decline (2016-2019): From around 2016, global FDI began to decline, due to factors such as increasing trade tensions, especially between the U.S. and China, and an overall slowdown in the global economy.
- 2020: The COVID-19 pandemic led to a dramatic drop in FDI. The onset of the pandemic and the associated global economic downturn resulted in significant disruptions to FDI. Many multinational enterprises delayed or canceled investment plans.

3. Commercial bank lending

- **Infrastructure and Public Goods:** Developing countries often require substantial investments in infrastructure, such as roads, electricity, water supply, and public health. Commercial bank loans to governments can finance these projects, which are crucial for economic development.
- **Private Sector Development:** Lending to enterprises, especially small and medium-sized enterprises (SMEs), can stimulate private sector growth. SMEs often face difficulties accessing finance, so bank loans can be a lifeline for them.
- **Job Creation:** Investments in both public projects and private enterprises can lead to job creation, helping reduce unemployment and underemployment in developing countries.

4. Export credits

- **Import of Essential Goods:** Developing countries often need to import capital goods, technology, or inputs not available domestically. Export credits make these imports affordable and accessible.
- **Technology Transfer:** Importing advanced technology or machinery can lead to technology transfer, helping domestic industries become more competitive and innovative.
- **Stimulating Trade:** By making it easier for countries to import, export credits can stimulate bilateral trade, which can lead to economic growth.
- **Currency Conservation:** Delayed payments mean that developing countries don't have to use their foreign exchange reserves immediately. This can be beneficial in managing their balance of payments.

B. Multinational Corporations (MNCs)

- MNCs: Corporations or enterprises that conduct and control productive activities in more than one country. FDI
Most in North America, Europe, Japan, South Korea, GM, Ford, Sony.
- Objective: Profit opportunities, unconcerned with issues such as poverty, inequality, unemployment.
- Countries received FDI from MNCs:
Most investment directed toward DCs. These countries offer a combination of advanced infrastructure, skilled labor, stable political and economic environments, and large consumer markets.

- Policies toward MNCs:

LDCs try to keep as much as possible benefits from FDI of MNCs.

1. Performance requirements

Use local materials.

Import only most – advanced technology.

2. Joint Ventures

Require MNCs to sell 51% of equity to local partners

3. Restrictions on profit repatriation

Example: Brazil at times limit profits sent by a firm
to a parent to 10%

4. Limit the tax holidays:

temporary reductions or eliminations of taxes, particularly corporate taxes, granted by governments. They are often used as incentives to attract foreign direct investment (FDI) or to promote the establishment and growth of new businesses within a country.

- Purpose of tax holidays: Attract FDI, Promote domestic industry growth
- Reasons to limit tax holidays: loss of revenue, short-term incentive, potential for misuse

Benefits of MNC:

- Consumers gain because of better quality, larger quantity products;
- Govt. receives more taxes;
- Domestic firms get technological knowledge;
- Increase employment;
- Transfer of capital
- Managerial capacity
- Access to world markets
- MNCs control 70% of trade

Arguments against MNCs:

- A. Widen wage differentials,
- B. Use capital intensive technology,
- C. Influence government policies in the directions unfavorable to development.
Tax rebates, cheap services.

C. Foreign Aid

(Chapter 14)

Any flow of capital to LDCs that meets two criteria:

- (1) its objective should be noncommercial from the point of view of the donor;
- (2) it should be characterized by concessional: low interest rate, longer repayment period.

1. Aid institutions

- United States Agency for International Development (USAID).
- The World Bank (International Bank for Reconstruction and Development) established in 1944 .

Borrow at mkt rates, lend at lower rates.

- The International Monetary Fund (IMF)
Monitor exchange rate and balance of payments policies and encourage trade.

- Regional development bank

Asian Infrastructure Investment Bank

2. Aid Instruments

- Project aid:

Infrastructure, dams, roads, rural development.

- Program loans:

Used to finance imports and support for balance of payment.

- Structural adjustment loans: (World Bank)

Support economic reforms.

- Food aid:

Commodities, grains.

- **Bilateral aid:**

- Given directly by one government to another.
- Direct Interaction
- Specific Purpose.
- Political and Economic Ties.

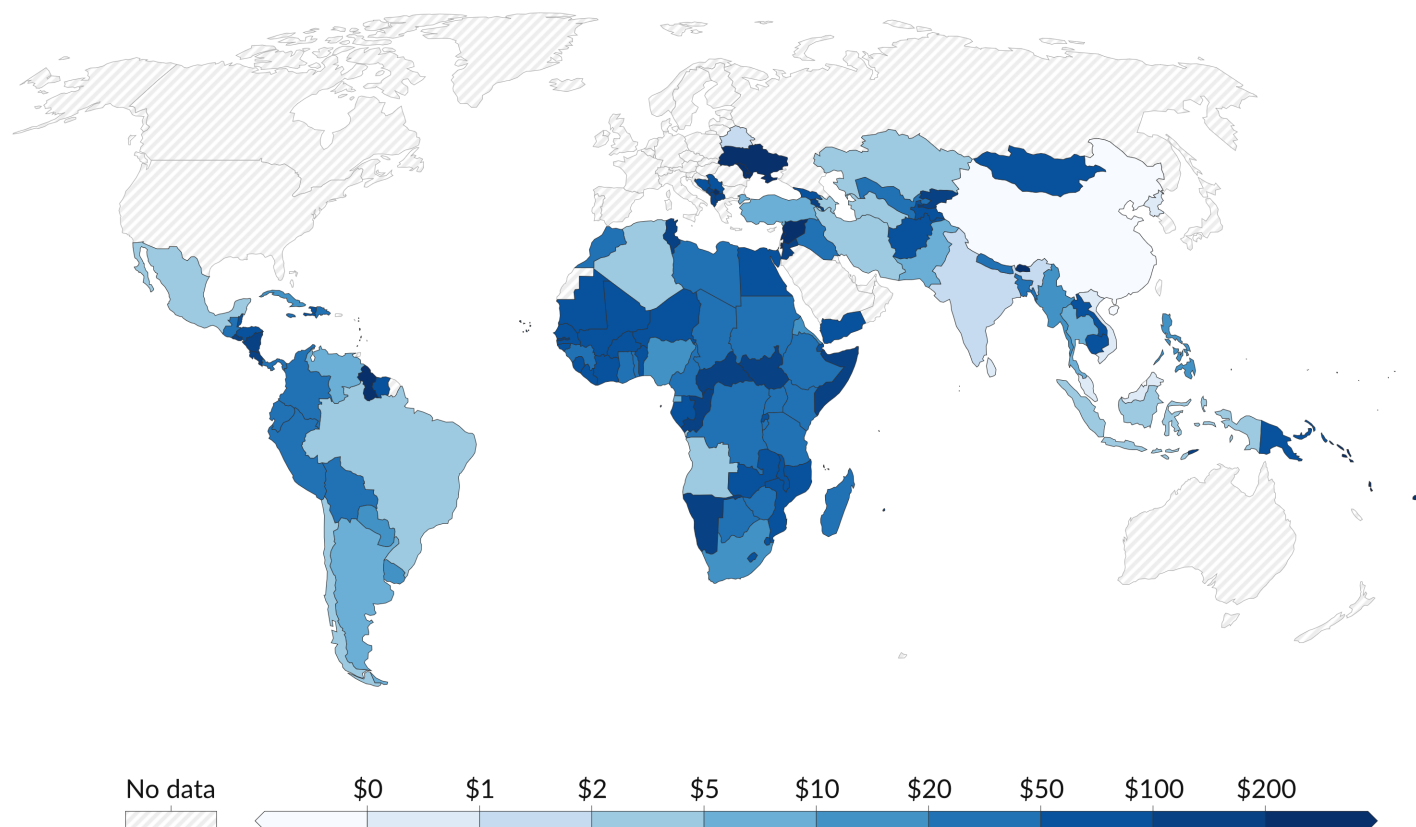
- **Multilateral aid:**

- Funds flow into international agencies which in turn grant or lend the funds to LDCs
- International Agencies.
- Broad Objectives.
- Diverse Sources

<i>Source:</i>	2005
<i>World Development Report 2008</i>	Official Aid per capita (\$)
World	17
Low Income Countries	17
Middle Income	15
low middle income	19
upper middle income	3
Low and Middle Income	20
East Asia and Pacific	5
Europe and Central Asia	12
Latin America and Caribbean	11
Middle East & North Africa	88
South Asia	6
Sub-Saharan Africa	43
High Income Countries	0
Selected Countries:	Russia 9
	Korea -1
	China 1
	India 2
	Congo, Republic 362
	Georgia 69
	Kuwait 1

Foreign aid received per capita, 2022

Net¹ official development assistance (ODA)², divided by population. This data is expressed in US dollars and adjusted for inflation.



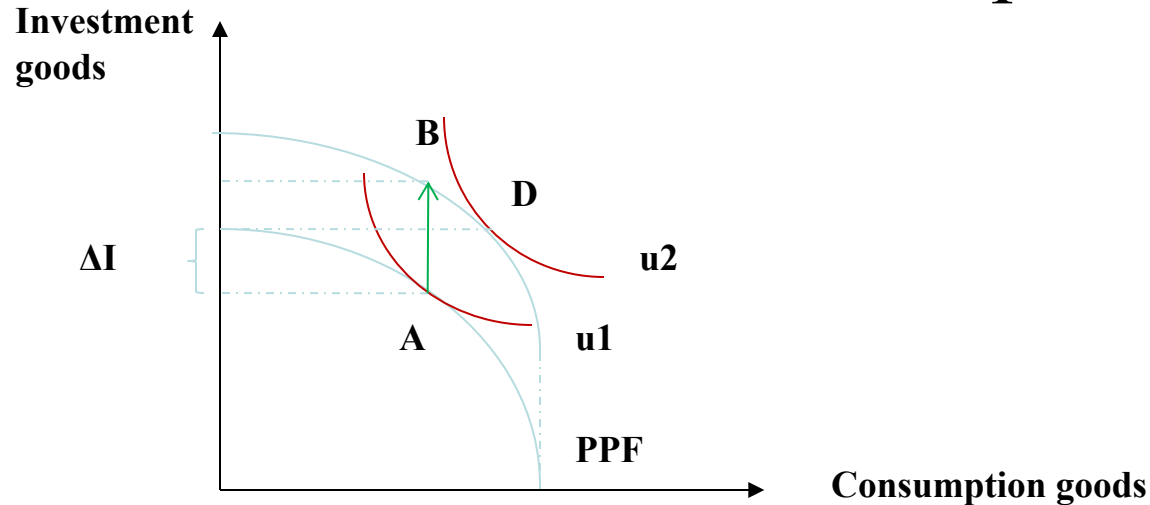
Data source: OECD (2024); Population based on various sources (2024)

OurWorldinData.org/foreign-aid | CC BY

1. Net (official development assistance): In official development assistance (ODA) statistics, "net" amounts mean that any money going out (like loan repayments or interest) has been subtracted from money coming in (like new grants or loans with favorable terms). Since more money can go out than comes in, net amounts can sometimes be negative.

2. Official Development Assistance: Financial flows from the Development Assistance Committee members (major donors in the OECD and EU institutions) to countries and territories on their list of recipients and to multilateral development institutions. These flows have to be provided by official agencies; serve the economic development and welfare of developing countries; and be concessional in character, with a grant element of at least 10-45 per cent, depending on the type of loan and the specific recipient.

3. Aid and Economic Development



- Foreign Aid = AB
- Actual Investment = ΔI
- Aid may be used to finance consumption
- There is not clear evidence that aid increases growth rate.

Aid affects development through:

1. Capital assistance (project aids)

- “Green Revolution” delivered new seeds and fertilizers to LDCs such as India & Indonesia.

Green Revolution refers to the transformation of agriculture started in 1945. Mexican government's request to establish an agricultural research station to develop more varieties of wheat to feed the growing population.

In 1943, Mexico imported half its wheat, but by 1956, self-sufficient; by 1964, Mexico exported half a million tons of wheat. The programs of agricultural research, extension, and infrastructural development were instigated and largely funded by the [Rockefeller Foundation](#) and the [Ford Foundation](#), and others.

2. Technical assistance

- Send experts to LDCs.

3. Conditionality

- Offer aid to friends & allies,
- Tie aid to imports,
- Some conditions may not help development.

4. Changing the minds of policymakers

- Send experts, scholars, officials to LDCs promote new ways of thinking

Cons of Aid:

- **Dependency:** Prolonged aid can lead to dependency, where a nation might not take initiatives on its own for development.
- **Misallocation:** Aid might not always be allocated efficiently. Corruption can divert aid away from those who need it most.
- **Debt Burden:** If aid is in the form of loans, it can lead to significant debt burdens.

- A significant example of corruption diverting foreign aid occurred in Zaire (now the Democratic Republic of the Congo) under the rule of Mobutu Sese Seko, who led the country from 1965 to 1997. During Mobutu's rule, Zaire received substantial foreign aid, primarily from Western nations, due to its strategic importance during the Cold War. However, Mobutu's administration was marred by extreme corruption, and he is believed to have diverted billions of dollars in aid and national resources into his personal accounts.





Criticism of Aid:

- **Political Motives:** Often, aid comes with strings attached, serving the political or strategic interests of the donor country.
- **Ineffectiveness:** Critics argue that despite trillions in aid over decades, many countries remain mired in poverty. Some believe that aid might even be detrimental by supporting corrupt regimes or by creating a dependency culture.

D. Foreign Debt

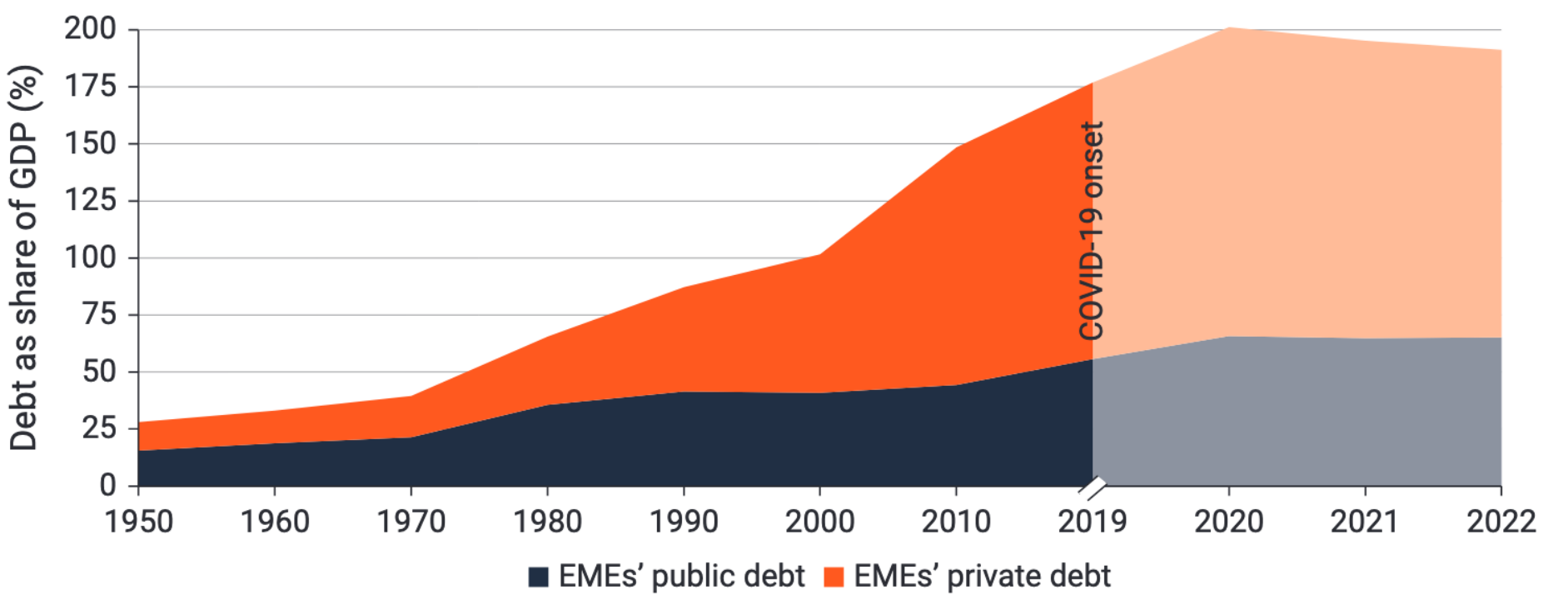
(Chapter 12)

All developing countries

	2000	2008
External debt stocks (\$ million)	2,152,234	3,718,539
External debt/GNI (%)	37.2	22.1
External debt/Exports (%)	122.2	58.7

Sources: Global Development Finance 2010

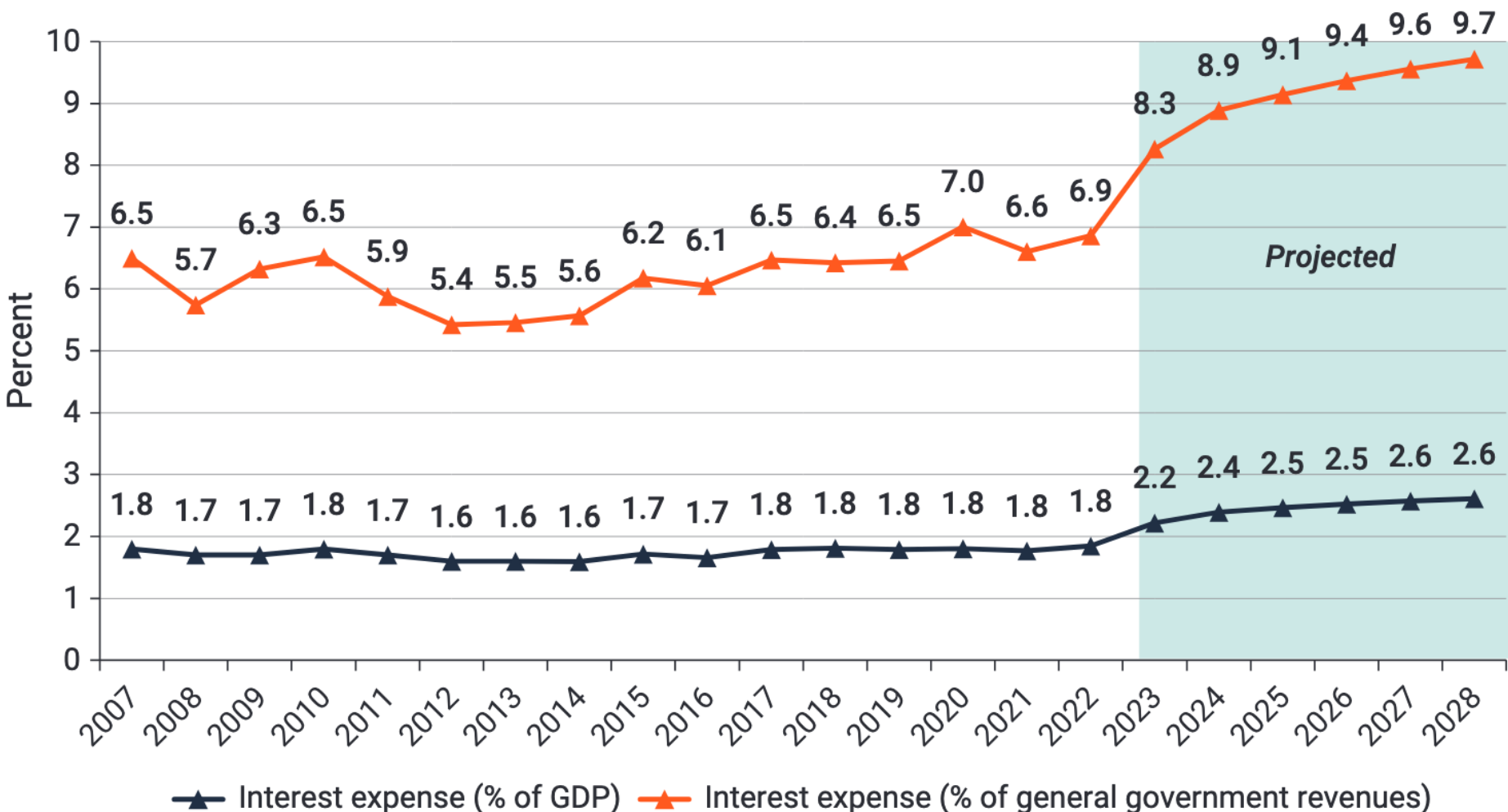
Figure 3.3 Most developing economies are more severely indebted than ever



Source: WDR 2024 team based on IMF (2023b); Kose and Ohnsorge (2024).

Note: The figure shows the total debt of emerging market economies (EMEs) in terms of weighted averages broken down into public debt and private debt. GDP = gross domestic product.

Figure 3.4 Debt service payments in emerging markets and middle-income countries may skyrocket as the cost of borrowing soars



Source: WDR 2024 team using data from Fiscal Monitor (dashboard), International Monetary Fund, Washington, DC, <https://data.imf.org/?sk=4be0c9cb-272a-4667-8892-34b582b21ba6>.

Note: GDP = gross domestic product

U.S. National Debt Stands at \$31.5 Trillion

Gross federal public debt in the United States
(in trillion U.S. dollars)



* as of May 2, 2023

Sources: Fed of St. Louis, Department of the Treasury



- Sustainable Debt:
 - the present value of future primary fiscal balances (government revenue minus expenditures, excluding interest payments) should be equal to the current stock of debt.
- Two-gap model : a conceptual framework used to understand the barriers to economic growth in developing countries.
 - The first gap refers to the difference between savings and investments,
 - The second gap pertains to the difference between exports and imports.
 - Both these gaps can be financed through external borrowings or foreign aid, leading to the accumulation of debt.
- $\Delta D = iD + M - E \quad (1)$
 - where D = debt, ΔD = change in debt, i = interest rate,
 - M = imports, E = export

Letting

$$g_E = \frac{\Delta D}{D} = \frac{\Delta M}{M} = \frac{\Delta E}{E}$$

$$\Rightarrow g_E D = \frac{\Delta D}{D} D = iD + M - E$$

$$\Rightarrow (g_E - i)D = M - E$$

$$\Rightarrow (g_E - i) \frac{D}{E} = \frac{M - E}{E}$$

$$\Rightarrow \frac{D}{E} = \left(\frac{M - E}{E}\right)/(g_E - i)$$

- If $g_E > i$, then M can be greater than E;
i.e. the country can borrow w/o raising Debt-export ratio.
D/E = 2 for developing countries in 1980s
D/E = 1.14 for developing countries in 2000
D/E = 0.59 for developing countries in 2008
- If $g_E < i$, then M must be smaller than E;
i.e. the country must begin to repay the debt (see eq. (1)).
This is what happened in the high-debt LDCs.
- If the growth rate of exports exceeds the interest rate, it implies that the country's earning capacity from exports is growing faster than the debt interest accumulation, which is favorable for debt sustainability.

- The investment-saving gap:

$$\Delta D = iD + I - S_d = iD + vY - sY$$

$$= iD + (v-s)Y, \quad v = I/Y$$

- Where $Y = \text{GNP}$, $v = \text{investment share}$, $s = \text{Marginal propensity to save}$,
 $s = S_d/Y$

- Letting $g_Y \equiv \frac{\Delta Y}{Y} = \frac{\Delta D}{D}$

$$g_Y D = iD + (V - S)Y$$

$$\Rightarrow (g_Y - i)D = (V - S)Y$$

$$\frac{D}{Y} = \frac{V - S}{g_Y - i}$$

$$\text{if } V - S = 3\%, \quad g_Y - i = 10\% - 7\% = 3\%$$

$$\text{then } \frac{D}{Y} = 1$$

- If $i > g_Y$, then $s > v$ to keep $\frac{D}{Y}$ constant,
- If $i < g_Y$, and $v > s$ to then $\frac{D}{Y} < 0$,
- If $i > g_Y$, & $\frac{D}{Y} > 0$, then $s > v$.

V. The Debt Crisis

1. History of International Lending

- In 19th century, U.K. 25-40% saving went to foreign countries.
- After World War I, U.K., France, Germany owed U.S. a large amount of money.

2. Debt crisis in 1930s

- In early 1930s, the world economy was in trouble. LDCs could not export enough to pay the debt.
- In 1931, Bolivia and then other Latin American Countries declared unable to pay the debt.
- Most LDCs could not get loans from DCs.

3. LDCs debt from 1945-1970

- Loans to governments, or loans guaranteed by governments.
- “Sovereign nations don’t go bankrupt”.

4. The debt increase in 1970s

- Low real interest rates in DCs, negative in U.S..
- OPEC put a large amount of money in DCs banks.

5. The crisis in 1980s

- In 1981, Mexico announced to be unable to pay the debt, followed by Brazil and Philippines.

6. The Asian Financial Crisis

- In 1997, debt crisis occurred in Thailand, then spread to S. Korea, Philippines, Indonesia, etc.

7. Greece Debt Crisis in 2009

Basic options confronted the creditors:

- Finance the Debt:
 - This means the creditor extends the payment terms, hoping the debtor country will be in a better financial position in the future and will eventually be able to repay.
 - From 1981 to 1989, this was the primary strategy adopted by U.S. policy.
- Forgive the Debt:
 - This involves reducing the amount owed by the debtor country to a level that they might be able to repay.
 - In 1989, U.S. Treasury Secretary Nicholas Brady announced a shift in U.S. policy, favoring a debt reduction strategy.